

TITLE:

ANHYDROUS ETHANOL

APPROVED IN: 19/05/2025
REVISION: 2

1. Identification

Name of substance or mixture (trade name): Anhydrous Fuel Grade Ethanol**Internal code of identification of the substance or mixture:** N/A**Contact phone:** (66) 3531-5494 / 0800 800 9595

Production Units					
Company Name	Inpasa agroindustria I S/A	Inpasa agroindustrial S/A	Inpasa agroindustrial S/A	Inpasa agroindustrial S/A	Inpasa agroindustrial S/A
Address	ROD BR 163 KM 817 CEP 78.559-899	ROD BR 163 KM 603 CEP 78.450-000	ROD BR 163 KM 243 Zona Rural CEP 79.849-899	ROD BR 060 KM 417 Zona Rural CEP 79170-000	ROD BR 230 KM 020 Zona Rural CEP 65800-000
Complement	Sinop – MT	Nova Mutum - MT	Dourados - MS	Sidrolândia - MS	Balsas - MA

2. Hazard identification

Product Rank:	• Flammable liquids – Category 2. • Severe eye injuries/eye irritation - Category 2A. • Germ cell mutation – Category 1B. • Toxicity to reproduction – Category 1A. • Toxicity to specific organs – Single exposure Category 3. • Toxicity to specific organs – Repeated exposure Category 1 and 2.
Classification system used:	• ABNT-NBR 14725-2:2009 Standard – corrected version 2:2010. • Globally Harmonized System for the Classification and Labeling of Chemicals, UN.
Other hazards that do not result in classification:	Vapors can form explosive mixtures in contact with air.

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Pictograms:	  
Word of warning:	DANGER
Danger Phrases:	• Extremely flammable liquid and vapors. • Causes severe eye irritation. • It can cause genetic defects. • May harm fertility or the fetus. • May cause irritation of the respiratory tract. • May cause drowsiness and vertigo. • May cause liver damage from repeated or prolonged exposure if ingested.
Precautionary Phrases:	• Keep away from heat, spark, open flame or hot surfaces. • DO NOT SMOKE. • IN CASE OF INHALATION: Remove the person to a ventilated place and keep him at rest in a position that does not hinder breathing. • IN CASE OF INGESTION: Contact a MEDICAL INFORMATION CENTER immediately. • IN CASE OF CONTACT WITH EYES: Rinse immediately with water for several minutes. In the case of contact lens wear, remove them if it is easy. Continue rinsing. • IF YOU ARE EXPOSED OR SUSPECTED EXPOSED: Contact a POISON CONTROL OR MEDICAL INFORMATION CENTER. • IN THE EVENT OF A FIRE: Use chemical powder, alcohol-resistant foam, carbon dioxide (CO2) and water mist.

3. Composition and ingredient information

SUBSTANCE	
Common chemical name or name	Ethanol

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technician:	
Synonym:	Ethyl alcohol; anhydrous alcohol; AEAC.
CAS Registration Number:	64-17-5 - min. 99,3 % (p/p)
Impurities that contribute to the danger:	This product does not contain any impurities that contribute to the hazard.

4. First aid measures

Inhalation:	Remove the victim to a ventilated place and keep him at rest. Monitor respiratory function. If the victim is breathing hard, provide oxygen. If necessary, apply artificial respiration. Seek medical attention. Take this FDS.
Skin contact:	Remove contaminated clothing and shoes. Wash exposed skin with a large amount of water for at least 15 minutes. Seek medical attention. Take this FDS.
Contact with eyes:	Rinse under running water for at least 15 minutes, keeping your eyelids open. Remove contact lenses when appropriate. Seek medical attention immediately. Take this FDS.
Ingestion:	Rinse the victim's mouth with plenty of water. DO NOT INDUCE VOMITING. Seek medical attention. Take this FDS.
Most important symptoms and effects, acute or late:	It can cause irritation to the skin and mucous membranes with redness and dryness. It can cause severe eye irritation with redness, tearing and pain. It can cause genetic defects. It can cause miscarriages, birth defects, and other developmental problems. It can cause depression of the central nervous system with dizziness, drowsiness, vertigo, headaches, motor incoordination and loss of consciousness. It can cause irritation to the respiratory and gastrointestinal tract with cough, sore throat, nausea, burning sensation, abdominal pain and diarrhea. It can cause liver damage with accumulation of fat in the liver and cirrhosis in case of chronic exposure by ingestion. It can cause damage to the central nervous system with tremors, irritability, loss of concentration and mental confusion.

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Notes to Doctor:	It can cause skin dryness after repeated and prolonged contact. Avoid contact with the product when helping the victim. Keep the victim at rest and warm. Do not give anything through your mouth to an unconscious person. Symptomatic treatment should include, above all, supportive measures such as correction of hydroelectrolytic and metabolic disorders, as well as respiratory assistance.
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5. Firefighting procedures

Means of extinguishing:	Suitable: Compatible with alcohol, water mist, chemical dust and carbon dioxide (CO₂) resistant foam. Not recommended: Water jets. Water directly onto the burning liquid.
Hazards specific to the mixture or substance:	When heated, it can release irritating and toxic gases. Containers can explode when heated. Risk of explosion indoors. Cool closed containers with sprayed water.
Fire team protection measures:	Self-contained respiratory protective equipment (SCBA) with positive pressure and full protective clothing. Containers and tanks involved in the fire must be cooled with water mist.

6. Control measures for spillage or leakage

Personal precautions for personnel who are not part of the emergency services:	Remove all ignition sources. Prevent sparks or flames. Do not smoke. Do not touch damaged containers or spilled material without wearing proper clothing. Avoid inhalation, contact with eyes and skin. Wear personal protective equipment as described in section 8.
For emergency service personnel:	Wear full PPE, with protective gloves made of PVC, natural rubber or nitrile, appropriate protective clothing and protective goggles or face shield against splashes. In case of large leaks, where the exposure is great, it is recommended to use a protective mask with a filter against organic vapors.

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Environmental Cautions:	Prevent the spilled product from encountering water and sewage network. Do not dispose it directly into the environment or sewer system. Dilution water from firefighting can cause contamination.
Methods and materials for containment:	Collect the spilled product and place it in proper containers. Adsorb the remaining product with dry sand, soil, vermiculite, or any other inert material. Place the adsorbed material in appropriate containers and remove them to a safe place. For final disposal, proceed according to section 13 of this FDS.
Differences in the action of large and small leaks:	There is no distinction between the actions of large and small leaks for this product.

7. Handling and storage

APPROPRIATE TECHNICAL MEASURES FOR HANDLING	
Precautions for safe handling:	Handle the product only in well-ventilated places or with general ventilation systems/suitable place. Avoid formation of vapors or mists. Avoid inhalation and contact with skin, eyes, and clothing. Avoid breathing vapors/mists from the product. Wear personal protective equipment as described in section 8.
Hygiene measures:	Do not eat, drink, or smoke while handling the product. Wash your hands thoroughly before eating, drinking, smoking, or going to the bathroom. Contaminated clothing should be changed and washed before reuse.
CONDITIONS FOR SECURE STORAGE, INCLUDING ANY INCOMPATIBILITY	
Fire and explosion prevention:	Keep away from heat, spark, open flame and hot surfaces. "Don't smoke."
Suitable conditions:	Keep the product in a cool, dry and well-ventilated place, away from heat sources and ignition. The storage place must contain a containment basin to retain the product, in case of leakage. Keep containers well closed and properly identified. The storage place must have an impermeable floor, free of combustible materials and with a containment dam to retain in case of leakage.
Packaging Materials:	Not specified.

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8. Exposure control and personal protection

CONTROL PARAMETERS			
Occupational exposure limits:	Component	TLV-STEL (ACGIH, 2012)	TL (NR 15,1978)
	Ethanol	1000 ppm	780 ppm
Biological indicators:	Not established.		
Engineering Control Measures:	Promotes mechanical ventilation and direct exhaust system to the outside environment. These measures help reduce exposure to the product. Maintain atmospheric concentrations of the constituents of the product below the indicated occupational exposure limits.		
PERSONAL PROTECTION MEASURES			
Eye protection:	Protective lenses or face shield against splashes.		
Skin and body protection:	Protective gloves (PVC or nitrile recommended) and product-resistant protective clothing (where there is a risk of splashes).		
Respiratory protection:	It is recommended to use a respirator with an organic vapor filter for average exposures above half of the TLV-TWA. In cases where exposure exceeds 3 times the TLV-TWA value, use a full-facepiece, self-supplied, self-contained respirator (SCBA) operated in positive pressure mode. Follow the guidelines of the Respiratory Prevention Program (PPR), 3rd ed. São Paulo: Fundacentro, 2002.		
Packaging Materials:	Not specified.		

9. Physical and chemical properties

Appearance (physical state, shape and color):	Clear and colorless liquid.
Odor and odor limit:	Distinctive. Odor limit: 180 ppm
pH:	6.0 – 8.0
Melting Point/Freezing Point:	-117°C

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Initial boiling point and boiling temperature range:	79 °C
Flash Point:	13°C (closed pot)
Evaporation Rate:	Unavailable.
Flammability (solid, gas):	Unavailable
Lower/upper flammability or explosiveness limit:	Higher: 19 % Low: 3.3%
Vapor pressure:	5.8 kPa a 20°C
Vapor density:	1.6 (air = 1)
Relative density:	0.8 (water at 4°C = 1)
Solubility(s):	Miscible in water, ethyl ether, acetone and chloroform. Soluble in benzene.
Partition coefficient – n-octanol/water:	Log kow: -0.32
Auto-ignition temperature:	363°C
Decomposition temperature:	Unavailable
Viscosity:	1.074 mPa.s a 20°C
Other information:	Unavailable

10. Stability and Reactivity

Stability and responsiveness:	Stable under usual handling and storage conditions. It does not undergo polymerization.
Possibility of dangerous reactions:	It reacts violently with strong oxidants such as nitric acid, silver nitrate, mercury nitrate or magnesium perchlorate with a risk of fire and explosion.
Conditions to be avoided:	High temperatures. Sources of ignition and contact with incompatible materials.
Incompatible materials:	Nitric acid, perchloric acid, permanganic acid, chromic anhydride, acetyl chloride, calcium hypochlorite, silver nitrate, mercury nitrate, hydrogen peroxide, bromine pentafluoride, perchlorates, silver oxide, ammonia and oxidants in general.

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Hazardous products of decomposition:

When combusting, it releases toxic and irritating vapors.

11. Toxicological information

Acute toxicity:	Not classified as toxic through oral ingestion. DL ₅₀ (rats): 7060 mg/kg
Skin corrosion/irritation:	It can cause skin irritation with redness and dryness.
Serious eye injuries/eye irritation:	It causes severe eye irritation with redness, tearing, and pain.
Respiratory or skin sensitization:	The product is not expected to cause respiratory or skin sensitization.
Germ cell mutation:	It can cause genetic defects. Information regarding: • Ethanol: Positive results for in vivo mutation assays involving mammalian germ and somatic cells.
Carcinogenicity:	The product is not expected to exhibit carcinogenicity. Information regarding: • Ethanol: Not classified as carcinogenic to humans (IARC).
Reproductive toxicity:	It can harm fertility or the fetus if ingested. Information regarding: • Ethanol: It can cause miscarriages, as well as birth defects and other developmental problems.
Specific organ toxicity – single exposure:	It can cause depression of the central nervous system with dizziness, drowsiness, vertigo, headaches, motor incoordination and loss of consciousness. It can cause irritation to the respiratory and gastrointestinal tract with cough, sore throat, nausea, burning sensation, abdominal pain and diarrhea.
Specific organ toxicity – repeated exposure:	It can cause damage to the central nervous system with tremors, irritability, loss of concentration and mental confusion. It can cause repeated and prolonged exposure to liver damage with accumulation of fat in the liver and cirrhosis in case of chronic exposure by ingestion. It can cause skin dryness after repeated and prolonged contact.

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Danger by aspiration: The product is not expected to present a hazard due to aspiration.

12. Ecological information

ENVIRONMENTAL EFFECTS, BEHAVIORS AND PRODUCT IMPACTS

Ecotoxicity:	Not classified as hazardous to aquatic organisms. LC50 (Salmo Gairdneri, 96h): 13000 mg/L
Persistence and degradability:	Rapid degradation and low persistence are expected.
Bioaccumulative potential:	It has low bioaccumulative potential in aquatic organisms. BCF: 3 Log kow: -0.32
Ground mobility:	Discharge.
Other adverse effects:	No other environmental effects are known for this product.

13. Considerations on destination

RECOMMENDED METHODS FOR TREATMENT AND DISPOSAL APPLIED TO:

Product:	They must be disposed of as hazardous waste in accordance with local legislation. Treatment and disposition should be evaluated specifically for each product. Federal, state and municipal legislation should be consulted, among them: Law No. 12,305, of August 2, 2010 (National Solid Waste Policy).
Leftover products:	Keep remains of the product in its original, closed packaging and inside metal drums, properly closed, in accordance with the applicable legislation. Disposal should be carried out as established for the product, recommending processing routes in cement plants and incineration.
Packing Used:	Never reuse empty packaging, as they may contain remains of the product and should be kept closed and sent to be destroyed in an appropriate place. In this case, it is recommended to send them to drum recovery routes or incineration.

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14. Transportation Information

NATIONAL AND INTERNATIONAL REGULATIONS	
Terrestrial	Resolução nº 5232 de 14 de dezembro de 2016 da Agência Nacional de Transportes Terrestres (ANTT), Aprova as Instruções Complementares ao Regulamento do Transporte Terrestre de Produtos Perigosos e suas modificações.
UN Number:	1170
Appropriate Name for Shipment:	ETHANOL (ETHYL ALCOHOL)
Risk class/main risk subclass:	3
Risk class/subsidiary risk subclass:	Not applicable
Risk Number:	33
Packing Group:	II
Waterway	DPC - Directorate of Ports and Coasts (Transport in Brazilian waters). Maritime Authority Standards (NORMAM) NORMAM 01/DPC: Vessels Used in Navigation on the Open Sea. NORMAM 02/DPC: Vessels Used in Inland Navigation. IMO – "International Maritime Organization".
INTERNATIONAL MARITIME DANGEROUS GOODS CODE (IMDG CODE).	
UN Number:	1170
Appropriate Name for Shipment:	ETHANOL (ETHYL ALCOHOL)
Risk class/main risk subclass:	3
Risk class/subsidiary risk subclass:	Not applicable
Packing Group:	II
EmS:	F-E, S-D
Danger to the environment:	The product is not considered a marine pollutant.
Aerial	ANAC - National Civil Aviation Agency - Resolution No. 129 of December 8, 2009. RBAC N°175 – (BRAZILIAN CIVIL AVIATION REGULATION) - TRANSPORT OF DANGEROUS ARTICLES IN CIVIL AIRCRAFT. IS N° 175-001 – SUPPLEMENTARY INSTRUCTION – IS ICAO – "International Civil Aviation Organization" – Doc 9284-NA/905. IATA - "International Air Transport Association".

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	Dangerous Goods Regulation (DGR).
UN Number:	1170
UN Number:	ETHANOL (ETHYL ALCOHOL)
Risk class/main risk subclass:	3
Risk class/subsidiary risk subclass:	Not applicable
Packing Group:	II

15. Regulatory Information

Regulations:	Federal Decree No. 2,657, of July 3, 1998. ABNT-NBR 14725:2023 Standard. Law No. 12,305, of August 2, 2010 (National Solid Waste Policy). Decree No. 7,404, of December 23, 2010. MTE Ordinance No. 704 of May 28, 2015 – Amends Regulatory Standard No. 26. Product subject to control and inspection by the Ministry of Justice – Federal Police Department – MJ/DPF, when it comes to import, export and re-export, and Prior Authorization from DPF is indispensable to carry out these operations. ANP Resolution 734/2018 Regulatory Standard No. 20c
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16. Other information

This SDS has been prepared based on current knowledge on the proper handling of the product and under normal conditions of use, according to the application specified on the packaging. Any other form of use of the product that involves its combination with other materials, in addition to forms of use other than those indicated, are the responsibility of the user.

It is cautioned that the handling of any chemical substance requires prior knowledge of its dangers by the user. In the workplace, it is up to the company that uses the product to promote the training of its employees and contractors regarding the possible risks arising from exposure to the chemical product.

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ACRONYMS:

ACGIH	American Conference of Governmental Industrial Hygienists
BCF	Bioconcentration Factor
CAS	Chemical Abstracts Service
CL50	Letal concentration 50%
DL50	Letal dosage 50%
IARC	International Agency for Research on Cancer
LEI - Limite de explosión inferior	Limite de explosividade inferior
LES - Limite de explosión superior	Limite de explosividade superior
NA – No aplicable	Não aplicável
STEL – Short Term Exposure Limit	Short Term Exposure Limit
TLV - Threshold Limit Value	Threshold Limit Value

BIBLIOGRAPHY:

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